
E-Mail jll0107@auburn.edu
Website <https://jllyle42.github.io/justinlyle/>

EMPLOYMENT

Auburn University Fall 2024-
Postdoctoral Fellow/Lecturer

Simons Laufer Mathematical Sciences Institute Spring 2024
Postdoctoral Fellow

University of Arkansas 2020-2023
Visiting Assistant Professor

EDUCATION

Ph.D. Mathematics 2015-2020
University of Kansas
Advisor: Hailong Dao
Thesis: Homological Properties of Structures in Commutative Algebra and Algebraic Combinatorics

M.A. Mathematics 2013-2015
Wake Forest University
Advisor: Frank Moore
Thesis: Noncommutative Complete Isolated Singularities

B.S. Mathematics (magna cum laude) 2009-2013
Berry College

PUBLISHED AND ACCEPTED WORKS

- [1] L. Ferraro, J. Lyle, The derived depth formula for modules of finite quasi-projective dimension, accepted to Bulletin of the London Math Society (2026), <https://arxiv.org/abs/2605.07004>.
- [2] J. Lyle, Endomorphism algebras over commutative rings and torsion in self tensor products, accepted to Acta Mathematica Vietnamica (2026), <https://arxiv.org/abs/2310.14134>.
- [3] E. Celikbas, O. Celikbas, J. Lyle, R. Takahashi, Y. Yao, On a generalization of Ulrich modules and its applications, Tohoku Mathematical Journal (2024), <https://arxiv.org/abs/2308.06606>.
- [4] J. Lyle, J. Montaña, K. Sather-Wagstaff, Exterior powers and Tor-persistence, Journal of Pure and Applied Algebra (2022), <https://www.sciencedirect.com/science/article/pii/S0022404921002310>
- [5] B. Holmes, J. Lyle, Rank selection and depth conditions for balanced simplicial complexes, Electronic Journal of Combinatorics (2021), <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v28i2p28>.
- [6] J. Lyle, J. Montaña, Extremal growth of Betti numbers and trivial vanishing of (co)homology, Transactions of the American Mathematical Society, Volume 373 (2020), pp. 7937-7958. <https://www.ams.org/journals/tran/0000-000-00/S0002-9947-2020-08189-6/>

- [7] T. Kobayashi, J. Lyle, R. Takahashi, On the number of indecomposable Cohen–Macaulay modules which are not locally free on the punctured spectrum, *Journal of Pure and Applied Algebra*, Volume 224 (2020) no. 7. <https://www.sciencedirect.com/science/article/pii/S0022404920300086>
- [8] H. Dao, J. Doolittle, J. Lyle, Minimal Cohen-Macaulay simplicial complexes, *SIAM J. Discrete Math.* (2020), 34(3), 1602-1608. (7 pages) <https://epubs.siam.org/doi/abs/10.1137/19M1275164>
- [9] H. Dao, J. Doolittle, K. Duna, B. Goeckner, B. Holmes, J. Lyle, Higher nerves of simplicial complexes, *Algebraic Combinatorics*, Volume 2 (2019) no. 5, pp. 803-813. https://alco.centre-mersenne.org/item/ALCO_2019__2_5_803_0
- [10] H. Dao, M. Eghbali, J. Lyle, Hom and Ext, revisited, *Journal of Algebra*, Volume 571 (2019), pp. 75-93. <https://doi.org/10.1016/j.jalgebra.2018.12.006>.

PREPRINTS AND SUBMITTED WORKS

- [1] L. Ferraro, J. Lyle, On End-semidualizing modules, preprint (2026), in preparation.
- [2] H. Lindo, J. Lyle, S. Maitra, Trace ideals of syzygies, preprint (2026), in preparation.
- [3] M. Brown, J. Lyle, Minimal models and polynomial growth of Betti numbers over dg-algebras, preprint (2026), <https://arxiv.org/abs/2605.11105>.
- [4] S. Dey, J. Lyle, Centers of endomorphism rings and reflexivity, preprint (2025), <https://arxiv.org/abs/2510.02210>.
- [5] L. Winther Christensen, A. Kekkou, J. Lyle, Z. Nason, A. Soto Levins, G -levels of perfect complexes and a Bass formula for levels, preprint (2025), <https://arxiv.org/abs/2507.10486>
- [6] K. Kimura, J. Lyle, A. Soto Levins, On the depth of tensor products over Cohen-Macaulay rings, preprint (2025), <https://arxiv.org/abs/2505.00441>.
- [7] J. Lyle, P. Mantero, Ordinary and symbolic powers of matroids via polarization, preprint (2025), <https://arxiv.org/abs/2505.17398>.
- [8] J. Lyle, S. Maitra, Annihilators of (co)homology and their influence on the trace ideal, preprint (2024), <https://arxiv.org/abs/2409.04686>.
- [9] J. Lyle, Generalized trace submodules and centers of endomorphism rings, preprint (2023), <https://arxiv.org/abs/2311.00220>.
- [10] K. Kimura, J. Lyle, Y. Otake, R. Takahashi, On the vanishing of Ext modules over a local unique factorization domain with an isolated singularity, preprint (2023), <https://arxiv.org/abs/2310.16599>.

TEACHING

Auburn University

- Math 1120 (Precalculus Algebra), Spring 2026
- Math 1617 (Honors Calculus I), Fall 2025
- Math 1680 (Calculus with Business), Fall 2024, Fall 2026

University of Arkansas

- Math 3083 (Linear Algebra), Spring 2023
- Math 2043 (Survey of Calculus), Fall 2021, Spring 2022, Fall 2022
- Math 2564 (Calculus II), Fall 2020, Spring 2021

University of Kansas

- Math 126 (Calculus II), Recitation Instructor, Spring 2019
- Math 104 (Precalculus), Instructor, Fall 2016, Fall 2018
- Math 115 (Applied Calculus), Instructor, Fall 2017, Spring 2018
- Math 127 (Calculus III), Recitation Instructor, Spring 2017

Wake Forest

- Math 121 (Linear Algebra), Teaching Assistant, Spring 2015
- Math 117 (Discrete Mathematics), Teaching Assistant, Fall 2014
- Math 113 (Multivariable Calculus), Teaching Assistant, Spring 2014
- Math 111 (Calculus I), Teaching Assistant, Fall 2013

AWARDS

Paul F. Conrad Scholarship <i>2019</i>	University of Kansas
Awarded annually to one or two mathematics graduate students for academic merit	
Chancellor's Fellowship <i>2015-2020</i>	University of Kansas
Department Summer Research Scholarship <i>2015-2019</i>	University of Kansas
Outstanding Graduate Student Award <i>2014</i>	Wake Forest
Barton Mathematics Award <i>2013</i>	Berry College

INVITED TALKS

AMS Spring Southeastern Sectional Meeting <i>Savannah, GA</i>	March 2026
"Annihilators of (co)homology and their influence on the trace ideal"	
AMS Fall Central Sectional Meeting <i>St. Louis, MO</i>	October 2025
"On the Depth of Tensor Products Over Cohen-Macaulay Rings"	
Syracuse University Algebra Seminar <i>Syracuse, NY</i>	April 2025
"Vanishing of Tor and Depth of Tensor Products Over Cohen-Macaulay Rings"	
UT Rio Grande Valley Algebra and Number Theory Seminar <i>Edinburgh, TX</i>	April 2025
"Vanishing of Tor and Depth of Tensor Products Over Cohen-Macaulay Rings"	
Commutative Algebra in the South (CATS) <i>Atlanta, GA</i>	April 2025
"Vanishing of Tor and Depth of Tensor Products Over Cohen-Macaulay Rings"	
Joint Math Meetings <i>Seattle, WA</i>	January 2025
"On the Vanishing of Tor and Depth of Tensor Products"	
AMS Spring Southeastern Sectional Meeting <i>San Francisco, CA</i>	May 2024
"Counterexamples for Several Open Problems on the Vanishing of Ext and Tor"	
AMS Spring Southeastern Sectional Meeting <i>San Francisco, CA</i>	May 2024
"Generalized Trace Submodules and Endomorphism Rings"	
SLMath Commutative Algebra Seminar <i>Berkeley, CA</i>	March 2024

“Counterexamples for Several Open Problems on the Vanishing of Ext and Tor” Joint Math Meetings <i>San Francisco, CA</i>	January 2024
“Endomorphism Algebras Over Commutative Rings and Torsion in Tensor Products” University of Utah Commutative Algebra Seminar <i>Salt Lake City, UT</i>	November 2023
“Counterexamples for Several Open Problems on the Vanishing of Ext and Tor” AMS Spring Southeastern Sectional Meeting <i>Cincinnati, OH</i>	March 2023
“Symbolic Powers of Squarefree Monomial Ideals, Revisited” AMS Spring Southeastern Sectional Meeting <i>Atlanta, GA</i>	March 2023
“Stable Theory for Complexes and the Auslander-Reiten Conjecture” West Virginia University Algebra Seminar <i>Virtual Session</i>	November 2022
“Stable Theory for Complexes and the Auslander-Reiten Conjecture” Auburn University Algebra Seminar <i>Auburn, AL</i>	November 2022
“Stable Theory for Complexes and the Auslander-Reiten Conjecture” Oklahoma State University Combinatorics and Commutative Algebra Seminar <i>Virtual Session</i>	April 2022
“Symbolic Powers of Squarefree Monomial Ideals, Revisited” University of Michigan Commutative Algebra Seminar <i>Virtual Session</i>	April 2022
“Generalizations of Ulrich Modules and Rigidity Theorems” Clemson University Algebra and Discrete Math Seminar <i>Virtual Session</i>	November 2021
“Recent progress on the Auslander-Reiten and Huneke-Weigand Conjectures” Iowa State University Commutative Algebra Seminar <i>Virtual Session</i>	April 2021
“Constructions in Combinatorial Commutative Algebra, Old and New” AMS Special Session on Commutative Algebra <i>Virtual Session</i>	April 2021
“Rank Selection and Depth Conditions for Squarefree Monomial Ideals” Wake Forest University Colloquium <i>Virtual Session</i>	March 2021
“Constructions in Combinatorial Commutative Algebra, Old and New” AMS Special Session on Homological Methods in Algebra <i>Virtual Session</i>	October 2020
“Tor-persistence and related conjectures” AMS Special Session on Commutative Algebra <i>Virtual Session</i>	April 2020
“Maximal Cohen-Macaulay modules that are not locally free on the punctured spectrum” University of Nebraska, Lincoln Commutative Algebra Seminar <i>Lincoln, NE</i>	March 2020
“Extremal growth of Betti numbers and rigidity of (co)homology” University of Nebraska, Lincoln Commutative Algebra Seminar <i>Lincoln, NE</i>	March 2020
“Maximal Cohen-Macaulay modules that are not locally free on the punctured spectrum” Purdue Commutative Algebra Seminar <i>West Lafayette, IN</i>	October 2019
“Generalizations of Ulrich Modules and Rigidity Theorems” AMS Special Session on Homological and Characteristic $p > 0$ Methods in Commutative Algebra <i>Madison, WI</i>	September 2019
“Generalizations of Ulrich Modules and Rigidity Theorems”	

KUMUNU Jr. <i>University of Nebraska, Lincoln NE</i> “Cohen-Macaulay Rings of Finite CM_+ -Representation Type”	March 2019
AMS Special Session on Commutative and Combinatorial Algebra <i>Auburn University, Auburn AL</i> “Cohen-Macaulay Rings of Finite CM_+ -Representation Type”	March 2019
AMS Special Session on Commutative Algebra <i>University of Arkansas, Fayetteville AR</i> “ ϵ -Ulrich Modules”	November 2018
AMS Special Session on Multiplicities and Volumes: An Interplay Among Algebra, Combinatorics, and Geometry <i>Ann Arbor, MI</i> “ ϵ -Ulrich Modules”	October 2018
KUMUNU Jr. <i>University of Nebraska, Lincoln NE</i> “Hom and Ext, Revisited”	April 2018

SERVICE

- Co-organizer of the 2027 iteration of the CATS (Commutative Algebra in the South) conference.
- Co-organizer of Special Session on “Topics in Commutative Algebra” at the AMS Spring 2026 Central Sectional Meeting.
- Auburn Math Carnival Exhibitor at the Destination STEM Expo, 2025: An expo for 6th-9th grade students in the Auburn community aimed at fostering interest in STEM fields.
- Co-organizer of the Career Development Series at SLMath, 2024
- Co-organizer of the Commutative Algebra Seminar at the University of Arkansas, 2020-2023.
- Co-creator and Co-organizer of CARES, 2020-2022: CARES is a regional seminar series uniting Arkansas, Kansas, Missouri, Nebraska, and Oklahoma State, and which focuses on expository talks given by graduate students. I served, along with Alessandra Costantini and Kyle Maddox, as a global advisor and organizer to the local organizers, who are senior graduate students at their respective institutions.
- Referee for:
 - Rocky Mountain J. Math
 - European Journal of Combinatorics
 - Communications in Algebra
 - AMS Graduate Series in Math
 - Journal of the Mathematical Society of Japan
 - Bulletin of the London Mathematical Society
 - Proceedings of the AMS
 - Journal of Algebra and its Applications
 - Journal of Commutative Algebra
 - Journal of Algebra
 - International Math Research Notices
 - Journal of Pure and Applied Algebra
 - Proceedings of the Edinburgh Mathematical Society